

ASSIGNMENT

$$ATOM = (A + TO + M)^2$$

Solve the cryptoarithmic problem.

- Q. A boy and girl have some pens. If a boy gave one pen to the girl, they would have same number of pens. If the girl gave one pen to the boy, the boy would have three times as many pen as the girl. How many pen do they have.
- Q. Prove that $11^{10} - 1$ is divisible by 100.
- Q. If $10^{1/10} > 2^{1/3}$.
- Q. What is the greatest amount of money that you could have with denominations of one paisa, five paisa, ten paisa and 25 paisa. Such that you still couldnot make up an exact 50 paisa.
- Q. Solve cryptoarithmic problem.

$$\begin{array}{r} \text{ONE} \\ + \text{ONE} \\ \hline \text{TWO} \end{array}$$

- Q. Positive integer beginning with one if kept continously writing, what digit of which number will be ~~retain~~ written by 3000th digit.
- Q. Which one is greater - 1000^{1000} or 1000^{999} .
- Binomial

Q. Sum:

$$1^2 + 2^2 + 3^2 + \dots$$

[Telescopy].

Q. If a government official offers to give you triangular plot of land with side 50 m, 100 m, 150 m. Why should you not be very happy or impressed getting that land. Prove that mathematically.

Q. ~~Derive the~~ Derive the largest and smallest two digit number which is seven times the sum of a digit.

Q. 5 red marbles and 4 blue marble are placed in a bag. What is the probability that 3 of them are blue.

Q. Before noon, it started snowing at constant rate (depth/hour). At noon or snow plough started to clear a snow at a constant rate (volume/hour). In the first hour, snow ploughs cleared two tractor of snow and in second hour, it clear one tractor of snow. find the time, when did the snow start snowing.

Q. Imagine five plane in 3D space are kept in general position. Now, find out how many region will this plane divide in the space.

Q. Examine the sequence:

$$1 = 1$$

$$1 - 4 = -(1 + 2)$$

$$1 - 4 - 9 = 1 + 2 + 3$$

$$1 - 4 - 9 - 16 = -(1 + 2 + 3 + 4)$$

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A ten year old child puts \$100 in bank. She intended to withdraw the money on her 21st birthday. Which one scheme is better for her.

- (i) An account with 5% interest compounded daily.
- (ii) An account with 5.1% interest compounded weekly.

If A, B, C, D are real numbers. Then show that:

$$\frac{(a^2+1)(b^2+1)(c^2+1)(d^2+1)}{abcd} \geq 76.$$

Examine the equation:

$$1 = 1$$

$$3 + 5 = 8$$

$$7 + 9 + 11 = 27$$

$$13 + 15 + 17 + 19 = 64$$

$$21 + 23 + \dots + 29 = 125$$

Determine the sequence.

- Q. Show that $\sqrt{2}$ and $\sqrt{7}$ are irrational numbers.
- Q. Explain why there are infinitely many prime numbers.
Hints: Prove by contradiction (Imp)
- Q. A certain K is multiple of 9. Add these digit together. If the result have more than 1 digit then add them together. Continue adding digit together until you have one digit. It would be 9. Explain why this so.
- Q. Show that Area of triangle is $A = \sqrt{s(s-a)(s-b)(s-c)}$
- Q. A ten foot pole is dropped in milling saw randomly cut into three pieces. What is probability three piece form a triangle.
- Q. Suppose that you have 9 liter container and 4 liter container, How can you put exactly 6 liter of water in container.
- Q. A right angled triangle has length L, m, n , note, L is not hypotenous and L and m are integer. Find L and m .
- Q. Draw a planer grid of 31 square wide and 17 square high. How many different non-trivial rectangle can be drawn using the line of grid to determine boundanes?
- Q. Prove Formula for Fibonacci Sequence.